

intensity on the LDR increases, the voltage at pin A0 rises. MOD1 converts this value into a digital signal between 0 and 1023 and maps it to a 0-100% range.

MOD1 serially transmits the values for all four parameters to the HC12 module via the Tx-Rx pins. The HC12 module modulates this data using a 434MHz carrier and transmits it wirelessly. Every 2-3 seconds, MOD1 reads updated sensor values and transmits them. The values are also sent to a PC/laptop via USB (COM port) for display on the Arduino IDE serial monitor.

In the receiver circuit, the HC12 module receives the transmitted data and serially sends it to MOD4, which displays it on the TFT LCD monitor. As the TFT LCD supports SPI communication, MOD4 sends data using SPI pins MOSI, MISO, and SCK. The display refreshes every 2-3 seconds with updated values. The range of the HC12 RF UART module exceeds 1.5km (line of sight), allowing data monitoring over long distances.

## Software

As the transmitter and receiver circuits require two different codes, the source code should be uploaded separately for each circuit in both MOD1 and MOD4. Please select the suitable board and ports before uploading the source code.

The transmitter source code performs the following tasks: It reads digital temperature and humidity values from the DHT11 sensor. It reads analogue voltage output from LDR and the soil moisture sensor, converting both outputs into digital values between 0-1023. It further converts values in this 0-1023 range into the 0-100 range to measure light and soil moisture level in a 0-100% range. It serially transmits all four parameter values using the HC12 module.

The transmitter source code uses

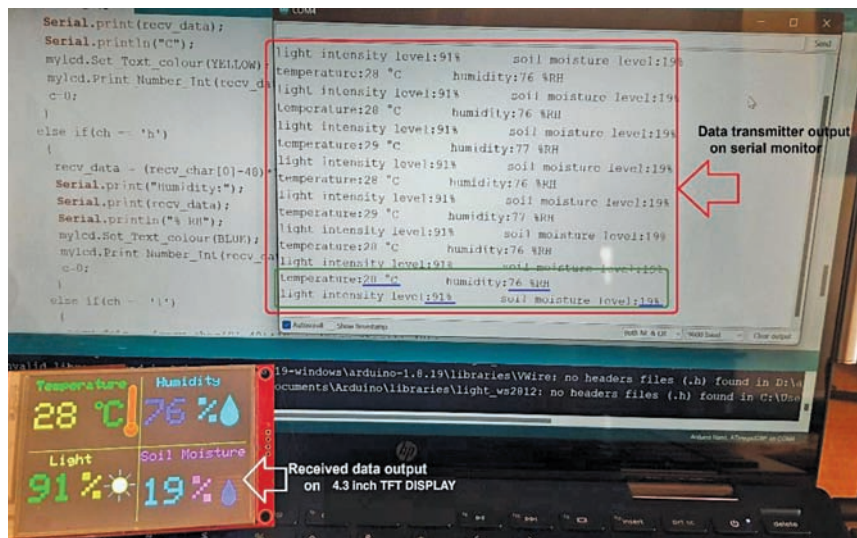


Fig. 8: Data output on the serial monitor and TFT LCD monitor

the DHT11 sensor library to initialise and configure the DHT11 sensor. It reads digital temperature and humidity values from the sensor. It uses SoftwareSerial to communicate with the HC12 module, sending data serially to the module using D4 (UART Tx) pin. This library transforms D3 and D4 pins of MOD1 as Rx and Tx pins for UART, respectively.

After serially receiving data from the HC12 module, the receiver source code displays temperature, humidity, light, and soil moisture level on the TFT LCD monitor.

It uses the LCDWIKI\_GUI.h library to display characters, numbers, shapes, and animation and create GUI on the TFT LCD monitor.

It uses the LCDWIKI\_SPI.h library to communicate with the TFT LCD display using the 4-wire SPI protocol. SoftwareSerial is used to communicate with the HC12 module, similar to that in the transmitter section. It receives data serially using D3 (UART Rx) pin from the HC12 module. This library makes D3 and D4 pins of MOD4 act as Rx and Tx pins for UART, respectively.

## Construction and testing

For constructing the long-range RDMS, follow the circuit diagrams of the transmitter (Fig. 4) and receiver (Fig. 6) along with the pin connections in Table 2. The prototype is shown in Fig. 7. Before assembling the circuits, upload the source codes to MOD1 and MOD4.

For testing, power both circuits. Fig. 8 shows the sensor data transmitted from the RDMS transmitter on the serial monitor as well as on the TFT LCD monitor. **EFY**

**EFY Note**  
The source code  
of this project  
is available  
for free download at  
[www.electronicsforu.com](http://www.electronicsforu.com)

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